Representations for Text

UW ECE 657A - General Topic

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Outline

1. Text Preprocessing
2. Distributional Hypothesis
3. Vector Space Models : TF-IDF
4. Word2Vec
5. GloVe

Feature Representations for Text Documents

Problems

- Overwhelming amount of text information.
- Search engines return linear ranked list of documents.
- Documents are of various topics yet interleaved.

Motivation

- Uncovering inherent groupings in large document sets.
- Easy navigation through large document sets.
- Easy navigation of search engine results.
- Many Useful Tasks:
  - Natural Language Processing, Document Classification, Text Understanding, Sentiment Analysis, …

Text Preprocessing Tasks

Preprocessing

- Stop-word Removal
  - (very common words like “the”, “and”)
- Stemming
  - (e.g. “computer” and “computing” become “comput”)

Feature Selection

- Information Gain (IG)
- Mutual Information (MI)
- Document Frequency (DF)
- X2 Statistic

Document Representation Model

"Standard" Approaches:

1. Vector Space Models (VSM)
2. Word Embedding based models (word2vec)
3. Global Vectors for Word Representation (GloVe)

Note: these methods were developed before recent almost revolutionary advances in NLP

The *goal* is to represent documents in a way that allows comparison of *semantics* and *meaning*, not just statistical correlations.

Three classes of VSM

1. Pair-pattern
   - rows= *carpenter:wood, …*
   - columns= *X cuts Y, …*
2. Term-document
3. Word-context
   - This is the most general case

Rely on the *distributional hypothesis*

P. D. Turney and P. Pantel (2010) "From Frequency to Meaning: Vector Space Models of Semantics", Volume 37, pages 141-188

The Distributional Hypothesis

Words with similar *context* have similar *meaning*.

- In the **term-document** setting, the context is the whole document.
- In the **word-context** setting, the context is some explicit context to the word
  - For **word2vec** in particular, it is a small neighbourhood around the current word

Term-Document Representation

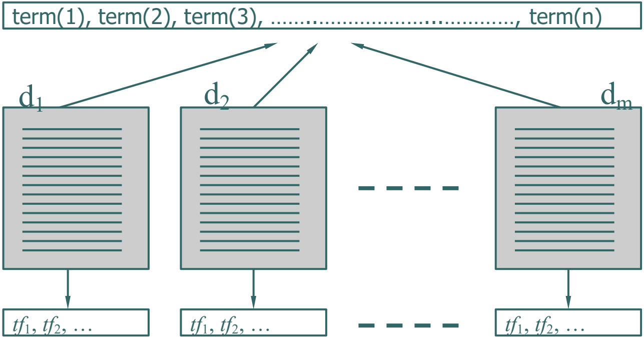
- A document represented by a *point in space*, a vector in vector space.
- Points close together are document that have similar meaning/content.

  - Representation:

$$d = \{tf_1, tf_2, \ldots, tf_n\}$$

where tf_i for $i = 1, \ldots, n$ is the **term frequency**

Term-Document Representation Model



Term Frequency Vectors
Also called
"Bag of Words"

Term-Document Representation Model

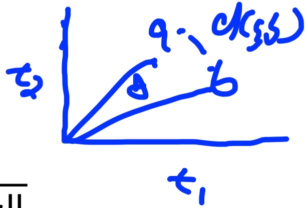
Term-Document matrix

Each word is represented by a vector of numbers based on their context

	t_1	t_2	t_3	....	t_n
$\mathbf{d}_1$	0.3	0	0.5	....	0
$\mathbf{d}_2$	0	0.1	0.4	....	0.2
⋮	⋮	⋮	⋮	....	⋮
$\mathbf{d}$	0.1	0	0	....	0

Similarity Measures

- Cosine Measure (common)



$$\cos(\boldsymbol{x}, \boldsymbol{y}) = \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{||\boldsymbol{x}||\,||\boldsymbol{y}||}$$

- Inverse of the Euclidean Distance

$$||x - y||_2$$

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Document Representation Model

Term Frequency Disadvantage (TF):

- A term appearing frequently in all documents has less discriminating power.

Document Frequency (DF):

- The *number of documents* in which a certain term appears.
- The *lower* the DF of a term, the *more* discriminating power it has.

TF-IDF

- **TF** is combined with the **Inverse Document Frequency (IDF)** to form the **Term Weight (TW):**

$$w_i = tf_i \times \log\left(\frac{N}{df}\right)_i$$

- High TF, Low DF → High TW
  - (i.e. high discrimination power)

Document Representation Model

Normalized tf-idf

- Now document length is taken into consideration

$$w_i = \frac{tf_i \times \log\left(\frac{N}{df_i}\right)}{\sqrt{\sum_j \left(tf_j \times \log\left(\frac{N}{df_i}\right)\right)}}$$

Normalized log(tf)-idf

- reduces the effect of large differences in term frequencies

$$w_i = \frac{\log(tf_i + 1) \times \log\left(\frac{N}{df_i}\right)}{\sqrt{\sum_j \left(\log(tf_i + 1) \times \log\left(\frac{N}{df_i}\right)\right)}}$$

Benefits and Drawbacks of TF-IDF

Benefits

- word sequence
- weighting terms by their importance (based on frequency)
- terms are independent and uncorrelated

Drawbacks

- ignoring term dependencies and correlations
- ignoring text structure, only co-occurrence in document
- ignoring ordering of the words in the document
  - IR research shows that word ordering is not important.
- ignoring grammar means it is language independent

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Word2Vec

- Word → Vector
- Goal is to learn a compact vector encoding of words in a document corpus (set of documents)
  - Simpler than an *auto-encoder*…why?
- Words in training data are represented by their *index* in the vocabulary
  - This is a form of *"one-hot encoding"*

Self-

- Learn a vector representation in a ~~semi~~-supervised way with a simplified neural network-type of structure
- Uses structure of sentence, relative position of words in sentence for similarity

Word2Vec – Two Main Approaches

- **Continuous Bag of Words (CBOW)**
  - Given a series of words, predict one in the middle
- **Skip-gram**
  - Given a single word, predict context words before and after

Word2Vec

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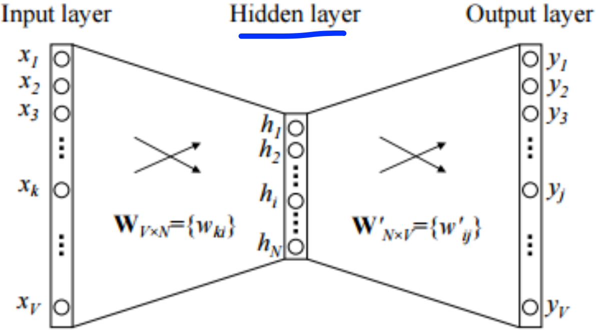
Vector of 1's
and
0's, only the
context
Word is 1



Distribution
Over outputs
of target word
Using a
softmax

Figure 1: A simple CBOW model with only one word in the context

Word2Vec

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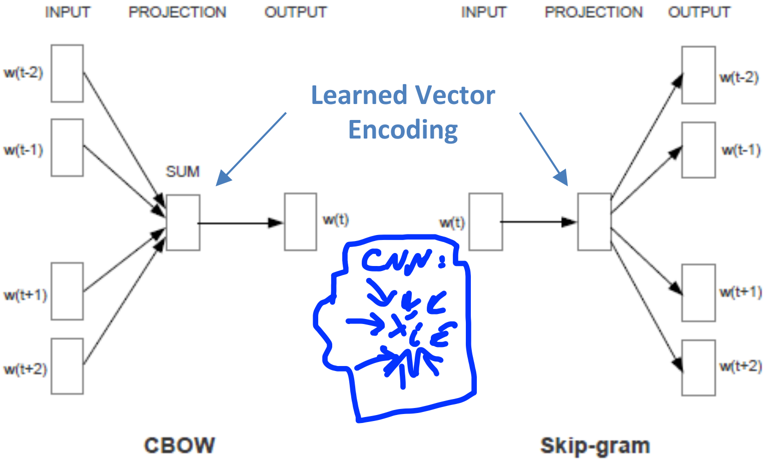
- Training data are series of words in order from document



Negative Sampling

Only some negative words considered. Otherwise there would be too many!

See : Zhen Yang. “Does Negative Sampling Matter? A Review with Insights into its Theory and Applications”

https://hyp.is/go?url=https%3A%2F%2Farxiv.org%2Fpdf%2F2402.17238v1&group=__world__

Word2Vec Training Process

- Parameters
  - (V) Vocabulary Size: 10,000? Size of corpus
  - (E) Embedding Size: 300? How many ‘features’?
  - (W) Window size: 5-50? How many words of context?
  - (S) Negative samples: 2-20? How many negative samples for each positive sample?
- Start with Two VxE Matrices:
  - Embedding Matrix
    - Initialized randomly
  - Context Matrix
    - Initialized randomly
- Blog Post Description: https://jalammar.github.io/illustrated-word2vec/

Word2Vec Training Process

- The Model will take a word as *input* and produce a predicted next more likely word as *output*
- To train the model…
- For each input word obtain the current wordvector for the
  - the true next output word
  - the multiple false next output words
  - compute the dot product of the the input with each of these outputs
  - Use the *sigmoid* function to produce a probability

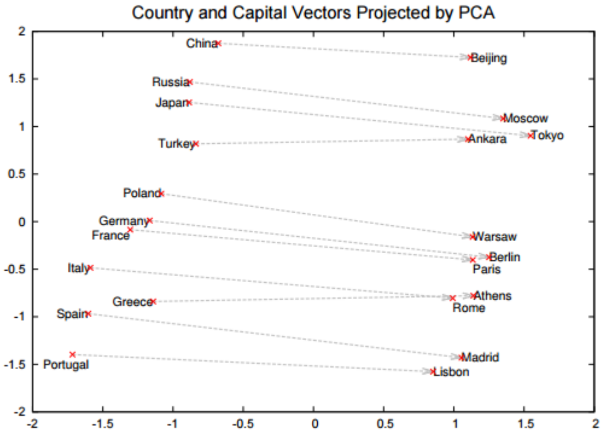
Word2Vec Training Process

- Compute the *error* that would occur from using each predicted output to predict the next word's probability
- Use the error to *update* the parameters of the sigmoid function for each input/output pairs using *logistic regression*

- Continue through all of the words in the text corpus to train the model and you end up with…

Vector arithmetic has semantic meaning.

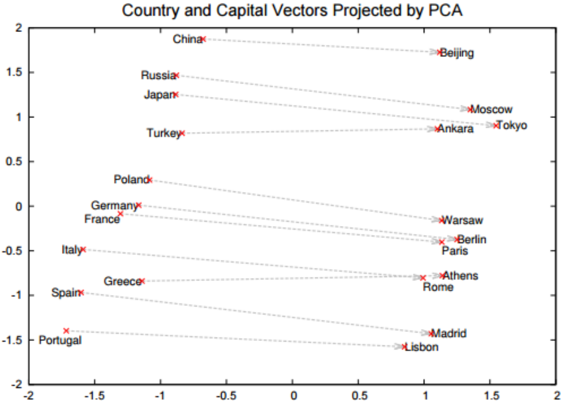
`vec(Berlin)-vec(Germany)+vec(China)~=vec(Beijing)`



Mikolov, Tomas, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. "Distributed Representations of Words and Phrases and Their Compositionality." In Advances in Neural Information Processing Systems 26 (NIPS 2013), 9, n.d.

Vector arithmetic has semantic meaning.

657A Steggal Lee Mar 12, 2025



Two-dimensional PCA projection of the 1000-dimensional Skip-gram vectors of countries and their capital cities. The figure illustrates ability of the model to automatically organize concepts and learn implicitly the relationships between them, as during the training we did not provide any supervised information about what a capital city means.

Example from Chemistry

This paper shows a ground-breaking demonstration of the power of vector encoding learning on scientific articles.

Tshitoyan, Vahe, John Dagdelen, Leigh Weston, Alexander Dunn, Ziqin Rong, Olga Kononova, Kristin A. Persson, Gerbrand Ceder, and Anubhav Jain. "Unsupervised Word Embeddings Capture Latent Knowledge from Materials Science Literature." Nature 571, no. 7763 (July 2019): 95–98. https://doi.org/10.1038/s41586-019-1335-8.

Try it Out Yourself

- https://remykarem.github.io/word2vec-demo/

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GloVe : Global Vectors for Word Representation

"GloVe is an **unsupervised** learning algorithm for obtaining vector representations for words. Training is performed on **aggregated global word-word co-occurrence statistics** from a corpus, and the resulting representations showcase interesting **linear substructures** of the **word vector space**."

- Created in 2014 by Jeffrey Pennington, Richard Socher, Christopher Manning at Stanford
  - https://nlp.stanford.edu/projects/glove/

GloVe : Global Vectors for Word Representation

- The Euclidean distance (or **cosine similarity**) between two **word vectors** provides an effective method for measuring the linguistic or semantic similarity of the corresponding words.
- Sometimes, the **nearest neighbors** according to this metric reveal rare but relevant words that lie outside an average human's vocabulary.

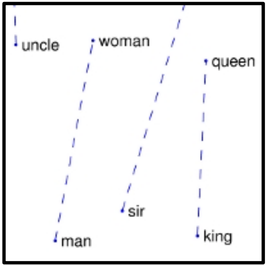
GloVe : Global Vectors for Word Representation

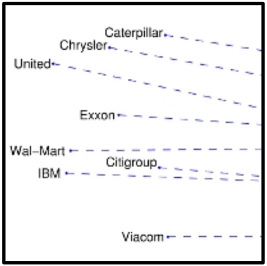
- GloVe is a log-bilinear model with a weighted least-squares objective.
- **Intuition :** the ratios of word-word co-occurrence probabilities have the potential for encoding some form of meaning.
- Some co-occurrence probabilities from a 6 billion word corpus for target words **ice** and **steam** :

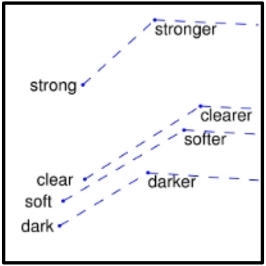
Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k\|ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k\|steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k\|ice)/P(k\|steam)$	8.9	8.5×10^{-2}	1.36	0.96

GloVe : Global Vectors for Word Representation

- Goal is to capture quantitatively the *meaning* of the difference bewteen *man* and *woman* in text
- It is necessary for a model to associate **more than a single number** to the word pair.
- **Solution:** use the **vector difference** between the two word vectors.
- GloVe is designed in order that such vector differences capture as much as possible the meaning specified by the juxtaposition of two words.







More Reading

- How to do negative sampling:
  - https://towardsdatascience.com/nlp-101-negative-sampling-and-glove-936c88f3bc68
- How to evaluate NLP models:
  - https://towardsdatascience.com/evaluation-of-an-nlp-model-latest-benchmarks-90fd8ce6fae5
- Leader Boards:
  - https://gluebenchmark.com/leaderboard
  - https://super.gluebenchmark.com/leaderboard/